

Week 1 Preview — What Is Computation? Strings, Input/Output, and Branching

What we'll learn this week

A computer is a machine that follows step-by-step instructions called **algorithms** — just like baking a cake from a box: mix, pour, bake, poke with a toothpick, and repeat until it comes out clean. We will learn how Python stores data (numbers, text, True/False), how programs talk to us (`input` / `print`), and how they choose between paths (`if` / `else`).

Terms to know

Term	One-line definition
algorithm	A sequence of simple steps, a flow of control deciding when each step runs, and a way to know when to stop
object & type	Programs manipulate data objects; each object has a type (<code>int</code> , <code>float</code> , <code>bool</code> , <code>str</code>) that decides what you can do with it
variable	A name bound to an object with <code>=</code> (assignment), e.g. <code>radius = 2.2</code> , so we can reuse the value
expression	Objects combined with operators (e.g. <code>3+2</code>); it evaluates to one value, which has a type
string (<code>str</code>)	A sequence of characters in quotes; you can index (<code>s[0]</code>), slice (<code>s[1:3]</code>), concatenate (<code>+</code>), and repeat (<code>*</code>) it
branching	<code>if</code> / <code>elif</code> / <code>else</code> structure: the program runs the first code block whose condition is <code>True</code>

Try this code before class

Taken from the Lecture 2 branching example (slide 42). **Predict the output before you run it.**

```
pset_time = 22
sleep_time = 8
if (pset_time + sleep_time) > 24:
    print("impossible!")
elif (pset_time + sleep_time) >= 24:
    print("full schedule!")
else:
    leftover = abs(24 - pset_time - sleep_time)
    print(leftover, "h of free time!")
print("end of day")
```

What will be printed? Is the last line always printed? Try changing `pset_time` to 15 and predict again before running.

Questions to think about

1. The lecture says "a computer will only do what you tell it to do." What everyday task do you follow so precisely that a machine could do it? Where would it break down?

2. Writing text in quotes ("3") and a number without quotes (3) are different types in Python. Why might it be useful — or dangerous — for a language to treat them differently?

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